Segregation from Background Noise as Outlier Detection

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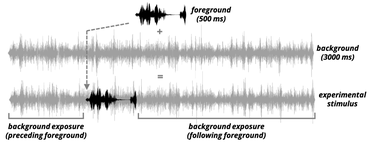
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Introduction

- The auditory system must segregate foreground events from noise in order to make sense of the everyday acoustic environment
- We explored whether listeners segregate sound events by estimating distributions over environmental background noise and registering outliers of these distributions as new events

Methods

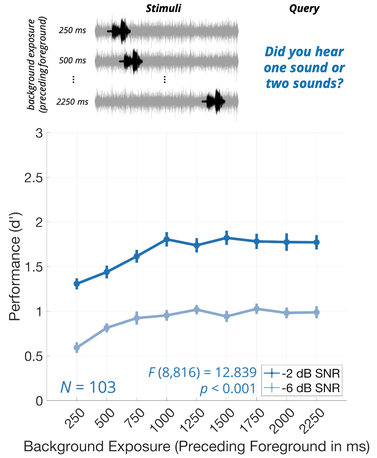
- Embedded foreground sounds in background noise (synthesized textures using real-world statistics)



- Manipulate timing of foreground and SNR (foreground rel. to noise)
- All data collected online via Amazon Mechanical Turk platform

Experiment 1: Foreground Detection

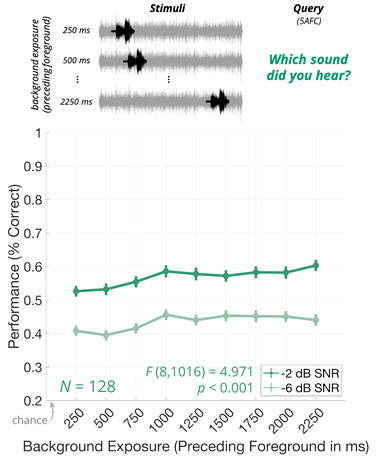
- Do listeners detect foreground sound events as outliers from a model of the background?



- Detection improves with time then plateaus, suggesting listeners benefit from collecting more samples from background

Experiment 2: Foreground Identification

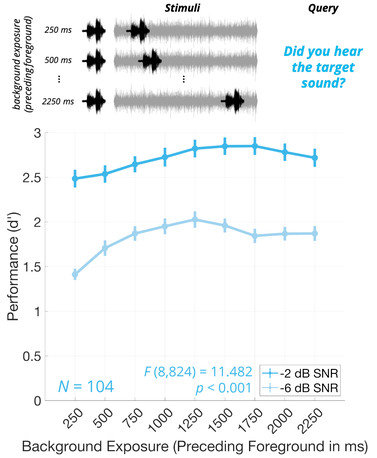
- Does this benefit of background exposure extend to sound source identification?



- Source recognition also benefits from increased background exposure

Experiment 3: Cued Detection

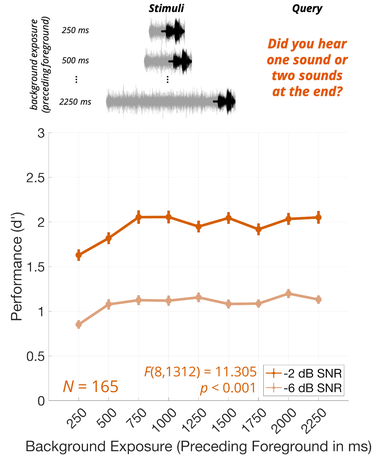
- Do listeners benefit from exposure to the background even when the foreground is known?



- The benefit of background exposure persists despite knowledge of the foreground

Experiment 4: Synchronous Offsets

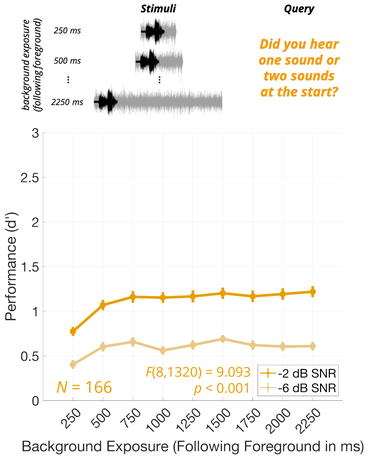
- Do listeners retrospectively detect foreground sound events under normal listening conditions?



- Suggests listeners do not use background following foreground offset to aid in detection

Experiment 5: Synchronous Onsets

- To what extent can listeners use the background to retrospectively detect foregrounds?



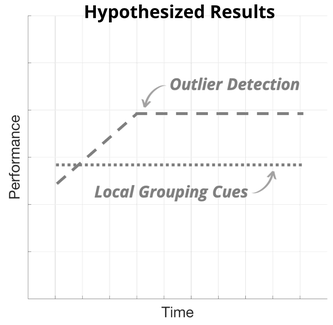
- Low overall performance
- Performance benefits from additional samples from the background but rapidly plateaus

Hypotheses

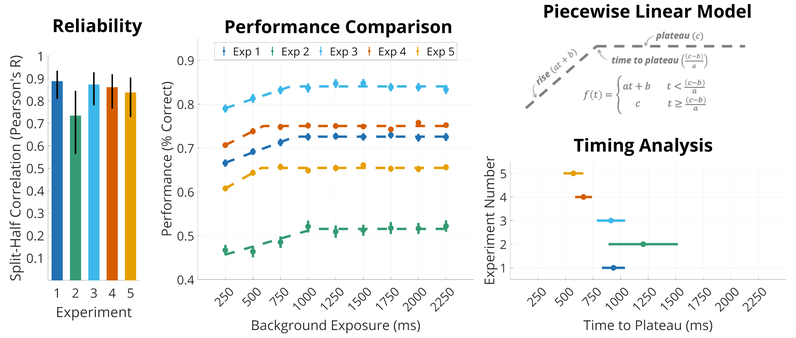
- Listeners detect acoustic events as outliers under a model of an environment's background texture
- Prediction: Ability to detect events will increase with longer exposure to background texture

vs

- Foreground sounds segregated via local grouping cues or features
- Prediction: Exposure to the background will have no effect on ability to detect events in noise



Results Summary



Conclusions

- Foreground detection performance improved over the first one-second exposure to the background
- Across experiments, the average improvement with time is approximately equivalent to a 2 dB improvement in SNR
- This perceptual mechanism is used across tasks and listening conditions
- Listeners can retrospectively detect foregrounds when asked to do so, but do not voluntarily retrospect even when it could be beneficial
- These results are consistent with the idea that listeners estimate the distribution of background noise and segregate sound events that are outliers from this distribution

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